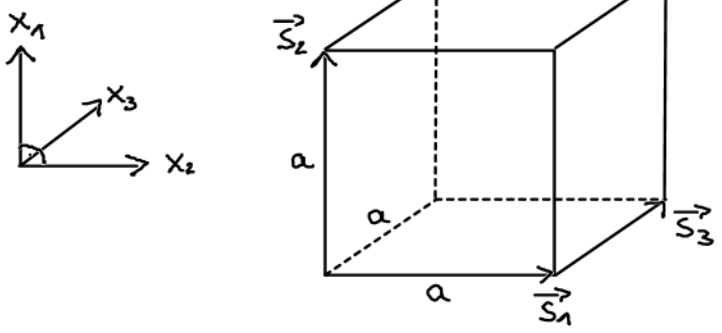
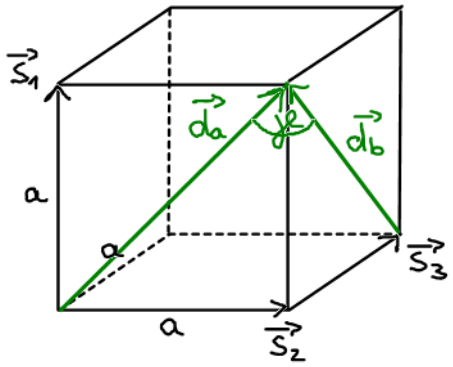
Lösung: Diagonalen eines Würfels



$$\vec{s}_1 = \begin{pmatrix} a \\ 0 \\ 0 \end{pmatrix} \qquad \vec{s}_2 = \begin{pmatrix} 0 \\ a \\ 0 \end{pmatrix} \qquad \vec{s}_3 = \begin{pmatrix} 0 \\ 0 \\ -a \end{pmatrix}$$

a) Winkel zwischen Flächendiagonalen



$$\vec{d}_a = \vec{s}_1 + \vec{s}_2 = \begin{pmatrix} a \\ 0 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ a \\ 0 \end{pmatrix} = \begin{pmatrix} a \\ a \\ 0 \end{pmatrix}$$

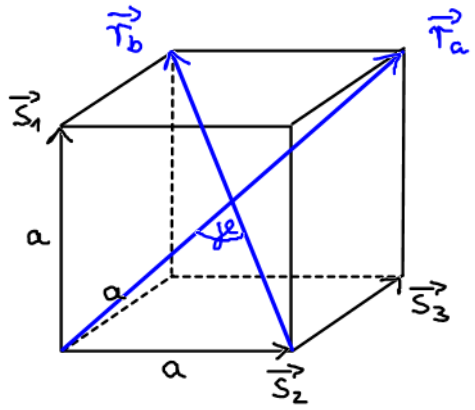
$$\vec{d}_b = \vec{s}_1 - \vec{s}_3 = \begin{pmatrix} a \\ 0 \\ 0 \end{pmatrix} - \begin{pmatrix} 0 \\ 0 \\ -a \end{pmatrix} = \begin{pmatrix} a \\ 0 \\ a \end{pmatrix}$$

$$\cos(\gamma) = \frac{\vec{d}_a \cdot \vec{d}_b}{|\vec{d}_a||\vec{d}_b|} = \frac{\begin{pmatrix} a \\ a \\ 0 \end{pmatrix}\begin{pmatrix} a \\ 0 \\ a \end{pmatrix}}{\sqrt{a^2+a^2+0^2}\cdot\sqrt{a^2+0^2+a^2}}$$

$$= \frac{a^2 + a\cdot 0 + 0\cdot a}{\sqrt{2a^2}\,\sqrt{2a^2}} = \frac{a^2}{2a^2} = \frac{1}{2}$$

$$\Rightarrow \quad \gamma = \arccos\left(\tfrac{1}{2}\right) = 60°$$

b) Winkel zwischen Volumendiagonalen



$$\vec{r}_a = \vec{s}_1 + \vec{s}_2 + \vec{s}_3 = \begin{pmatrix} a \\ a \\ -a \end{pmatrix}$$

$$\vec{r}_b = \vec{s}_1 - \vec{s}_2 + \vec{s}_3 = \begin{pmatrix} a \\ -a \\ -a \end{pmatrix}$$

$$\cos(\varphi) = \frac{\vec{r}_a \cdot \vec{r}_b}{|\vec{r}_a||\vec{r}_b|} = \frac{\begin{pmatrix} a \\ a \\ -a \end{pmatrix}\begin{pmatrix} a \\ -a \\ -a \end{pmatrix}}{\sqrt{a^2+(-a)^2+(-a^2)} \cdot \sqrt{a^2+(-a)^2+(-a)^2}}$$

$$= \frac{a^2 - a^2 + a^2}{\sqrt{3a^2}\sqrt{3a^2}} = \frac{a^2}{3a^2} = \underline{\underline{1/3}}$$

$$\Rightarrow \quad \underline{\varphi = \arccos(1/3) = 70{,}53°}$$